

The Price of Control: Economic Costs and Strategic Logic of Technology Export Restrictions

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Export controls on advanced technology occupy an unusual position in the economic policy toolkit. Unlike tariffs, which raise the price of imports while generating government revenue, or sanctions, which sever financial relationships, export controls impose costs on the exporting firm without generating any domestic fiscal benefit. The seller forgoes a transaction. The buyer seeks alternatives. The controlling government bears the political cost of restricting its own industry in exchange for a strategic benefit that may materialize only years later, if at all. Standard trade economics provides limited guidance here because the instruments are not welfare-maximizing in the usual sense; they are instruments of strategic denial whose value is explicitly non-economic.

The United States government's October 2022 restrictions on semiconductor exports to China, and their October 2023 successor rules, are the most consequential application of export controls in the technology sector since the Cold War-era Coordinating Committee for Multilateral Export Controls (CoCom). Evaluating them requires a framework that takes the strategic logic seriously while accounting for the real economic costs they impose.

A Brief History of Technology Export Controls

The United States has maintained formal export control mechanisms since the Export Control Act of 1949, which authorized the executive branch to restrict exports of goods deemed critical to national security. The primary Cold War vehicle was CoCom, a multilateral arrangement among NATO members and Japan that maintained a list of technologies denied to the Soviet bloc. CoCom's defining feature was allied coordination: restrictions were only effective if the target country could not obtain the controlled goods from a third party, so maintaining allied participation was as important as maintaining the control list itself.

CoCom dissolved in 1994 following the Soviet Union's collapse. Its successor, the Wassenaar Arrangement on Export Controls for Conventional Arms and Dual-Use Goods and Technologies, adopted in 1996 and now encompassing 42 participating states, takes a consensus-based approach in which member states exchange information about denials but retain national discretion over approvals.¹ Wassenaar's structure is considerably weaker than CoCom's: a single member state's decision to approve a transaction is not blocked by others' objections. This structural difference matters for evaluating the effectiveness of controls that require allied coordination to function.

The October 2022 Semiconductor Rules

On October 7, 2022, the U.S. Bureau of Industry and Security issued an interim final rule that fundamentally restructured export controls on advanced semiconductors and related manufacturing equipment.² The rules imposed new licensing requirements on exports to China of chips exceeding specified performance thresholds, as well as on the equipment, software, and components used to manufacture such chips at advanced process nodes. Crucially, the rules also restricted U.S. persons -- including employees of foreign firms -- from supporting the development or production of advanced chips in China without a license. This

extraterritorial provision targeted the human capital dimension of semiconductor development in addition to the hardware dimension.

The October 2023 rules updated the framework with a revised performance density metric designed to close loopholes that Chinese firms had begun to exploit, and added restrictions on additional chip architectures and countries of concern.³

The breadth of the 2022 rules was unusual. Previous semiconductor export controls had targeted specific products (the A100 and H100 GPUs were restricted in August 2022, before the broader October rule). The October rules moved to control the means of production -- the lithography equipment, deposition tools, and electronic design automation software needed to manufacture advanced chips -- rather than only the products themselves. This represented a shift in strategic logic from denying capability at the product level to denying it at the process level.

Economic Costs to U.S. Firms

The costs to U.S. semiconductor firms were material and publicly disclosed. Nvidia disclosed in its Q3 fiscal year 2024 earnings filings that China and affected destinations had accounted for approximately 20 to 25 percent of Data Center revenue in the preceding quarters.⁴ Following the October 2023 rule updates, the company stated it expected sales to those destinations to decline significantly, though it anticipated that growth in other regions would offset the loss. Applied Materials, Lam Research, and KLA -- the three largest U.S. semiconductor equipment companies -- collectively derived between 25 and 35 percent of their revenues from Chinese customers in the period preceding the controls.⁵

These are direct revenue losses from businesses that were legal and profitable prior to the controls. They are not deadweight losses in the welfare-economic sense; the transactions simply do not occur, and both parties lose the gains from trade. From the perspective of the controlling government, the costs are a payment made in exchange for the strategic benefit of delaying or degrading a competitor's access to advanced manufacturing capability.

The incidence of these costs is concentrated in firms with large China exposure. Smaller equipment and materials companies with higher proportional China revenue faced more acute impacts; larger diversified firms absorbed the revenue loss more easily against a backdrop of strong demand in the rest of the world, where AI-driven investment in advanced chip production was accelerating simultaneously.

Strategic Effectiveness: The Economics Literature

The economics literature on export controls offers a sobering baseline. The general finding from empirical studies of Cold War-era controls is that they imposed costs on target countries but rarely achieved their stated objective of preventing the controlled capability from eventually diffusing.⁶ Soviet bloc countries developed, through indigenous effort and third-party acquisition, many of the technologies that CoCom sought to deny, typically with a lag of several years to over a decade. The controls bought time rather than permanent denial.

The buying of time has independent strategic value, particularly in technology domains characterized by rapid progress. A five-year delay in an adversary's acquisition of a manufacturing capability is not the same as permanent denial, but it may allow the controlling country to maintain a lead of two to three technology generations -- a meaningful advantage in a sector where performance roughly doubles every two to three years. The question for the 2022 semiconductor rules is therefore not whether they can permanently prevent

China from developing advanced semiconductor manufacturing, but whether the delay they impose is long enough and the capability gap they maintain large enough to produce durable strategic benefit.

The Huawei Case as Partial Evidence

Huawei's trajectory following entity list placement in May 2019 -- which preceded the broader October 2022 rules -- provides partial evidence about the effectiveness of denial at the firm level. Before 2019, Huawei fabbed its flagship Kirin system-on-chip at TSMC using the most advanced available process nodes. Entity list placement cut off TSMC access in 2020. By 2023, Huawei had fabricated the Kirin 9000S on SMIC's DUV-based 7nm process, demonstrating that denial had not eliminated capability but had imposed a significant performance and cost penalty. The chip's performance lagged TSMC-fabricated equivalents by one to two technology generations, and its production yield was substantially lower.⁷

This pattern is consistent with the Cold War literature: controls slow progress and impose cost without permanently preventing capability development. The degree of slowing and the magnitude of cost depend on the availability of alternative suppliers, the substitutability of controlled inputs, and the indigenous engineering capacity of the target country.

Allied Coordination as the Critical Variable

The 2022 rules included a significant multilateral dimension. The United States successfully negotiated coordinated restrictions on ASML's export of EUV lithography equipment from the Netherlands and on certain Nikon and Tokyo Electron equipment from Japan. This coordination was consequential: ASML is the sole global supplier of EUV systems, and a unilateral U.S. restriction on its own (non-existent) EUV exports would have accomplished nothing. The restriction works because the Netherlands agreed to enforce it.

The fragility of this coordination is the central variable for long-run effectiveness. CoCom eventually failed not because the technology it controlled became unimportant, but because the interests of member states diverged as commercial opportunities in the formerly controlled markets grew. The economic pressure on ASML -- which cannot sell its most advanced products to a major and growing market -- is permanent and escalating. Sustaining allied commitment to restrictions that impose large and visible costs on allied firms in exchange for diffuse strategic benefits is a recurring challenge in multilateral technology control regimes, and the semiconductor case is no exception.

References

¹ Wassenaar Arrangement on Export Controls for Conventional Arms and Dual-Use Goods and Technologies, *Introduction*, <https://www.wassenaar.org/the-wassenaar-arrangement/> (last accessed 2024).

² U.S. Department of Commerce, Bureau of Industry and Security, "Implementation of Additional Export Controls: Certain Advanced Computing and Semiconductor Manufacturing Items; Supercomputer and Semiconductor End Use; Entity List Modification," 87 Fed. Reg. 62186 (October 13, 2022).

³ U.S. Department of Commerce, Bureau of Industry and Security, "Export Controls for Advanced Computing and Semiconductor Manufacturing Items to Countries of Concern; Supercomputer and Semiconductor End Use; Updates and Corrections," 88 Fed. Reg. 73424 (October 25, 2023).

⁴ Nvidia Corporation, Q3 FY2024 CFO Commentary, filed with the U.S. Securities and Exchange Commission (November 21, 2023). The 20-25 percent figure is drawn directly from Nvidia's investor disclosures for the

period preceding the October 2023 rule update.

⁵ Applied Materials, Lam Research, and KLA annual reports and investor presentations, fiscal years 2021-2022. China revenue share figures are publicly disclosed in SEC filings.

⁶ Michael Mastanduno, *Economic Containment: CoCom and the Politics of East-West Trade* (Ithaca: Cornell University Press, 1992); Gary Bertsch and Steven Elliott-Gower, eds., *Export Controls in Transition: Perspectives, Problems, and Prospects* (Durham: Duke University Press, 1992). For a more recent synthesis, see Chad Bown, "The U.S.-China Trade War and Phase One Agreement," *Journal of Policy Modeling* 43, no. 4 (2021): 805-843.

⁷ TechInsights, "Huawei Kirin 9000S Teardown Analysis" (2023); Reuters, cited in Financial Times and EDN, reporting on SMIC 7nm yield rates in 2023-2024.